

DP-MLM-X

Risk-Aware Differential Privacy for Text Anonymization

Yamaç Eren Ay

Faculty of Electrical Engineering and Computer Science

2026-07-02

Contents

Motivation	3
DP-MLM-X: Risk-Aware Anonymization	7
Results	14
Takeaway	20

Motivation

Original biography

John Smith was born on **12 March 1987** in **Munich**.
He worked as a **senior software engineer** at **Deutsche Bahn** from 2014 to 2019 and resided in Berlin.

Red = high re-ID risk · Orange = medium · Grey = low

After standard name / date masking

[PERSON] was born on **[DATE]** in **[LOC]**.
He worked as a **senior software engineer** at **Deutsche Bahn** from 2014 to 2019 and resided in Berlin.

A re-identification model still ranks him #1 of 2 419 candidates.

Occupation and employer remain untouched.

Uniform DP-MLM — $\epsilon = 10$

Maria Garcia was born *during September 1991 near Hamburg*.
He worked as *the junior analyst over Lufthansa*
from *2016* to 2019 and *resided in Berlin*.

- 10 tokens rewritten (45 % of content words)
- Common words distorted: “a” → “the”, “in” → “near”
- Text becomes unnatural — utility collapses

Risk-aware DP-MLM-X — same $\epsilon = 10$

Thomas Weber was born on *3 September 1990 in Stuttgart*.
He worked as a *senior software engineer at Siemens*
from 2014 to 2019 and *resided in Berlin*.

- 4 tokens rewritten (18 % of content words)
- Structure and low-risk words preserved
- Text stays readable — utility maintained

Same budget. Stronger privacy. Less distortion.

DP-MLM-X: Risk-Aware Anonymization

SHAP: A Re-Identification Risk Flashlight

Antonio Barbalonga · Baroque painter

[PERSON] or [LOCATION] (1600 – 2 [DATE_TIME]), also called [PERSON], was an [NRP] painter of the Baroque period. He was a member of the noble family of the [PERSON], born at [LOCATION], and was there instructed in painting by [PERSON]. He went to [LOCATION], where he became a pupil of [PERSON], whose style he imitated with great skill. **Barbalonga** executed a great number of paintings for churches, his chief work being the Conversion of St. Paul for the convent church of [LOCATION] at [LOCATION]. He painted in [LOCATION] for both the Church of the Theatines (San Silvestro [PERSON]) for the church of Sant' [PERSON], and for the church of [LOCATION] in [LOCATION] in [LOCATION]. In

William Oliver · physician / Madron

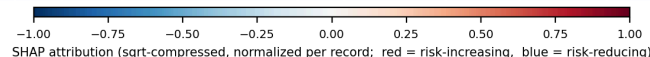
[PERSON] ([DATE_TIME] [O.S. [DATE_TIME]] [PHONE_NUMBER] March 1764) was an [NRP] physician and philanthropist, and inventor of the Bath Oliver. He was born at [LOCATION], [LOCATION], and baptised on [DATE_TIME], described as the son of [PERSON] the owner of the Trevarno Estate. His family, originally seated at [PERSON] in Sithney, resided afterwards in [LOCATION], and the estate of [LOCATION] in **Madron**, which belonged to him, was sold in [DATE_TIME] after his death. When he decided to erect a monument in Sithney churchyard to the memory of his parents, [PERSON] wrote the epitaph and drew the design of the pillar. He was admitted a pensioner of Pembroke College, [LOCATION] on

Yosef Tekoah · Ben-Gurion diplomat

[PERSON] (born [PERSON]; [DATE_TIME] – [DATE_TIME]) was a senior [NRP] diplomat and an [NRP] doctor and the President of the **Ben-Gurion University** of the Negev (1975–1981). [PERSON] was born in [LOCATION], [LOCATION] as [PERSON]. At [DATE_TIME] he emigrated with his family to [LOCATION], due to the rise of fascism in his homeland. Some time after the Fall of [LOCATION] to the Imperial Japanese Army, [PERSON]'s family moved to [LOCATION] for financial purposes. He had a Doctorate in international relations from Harvard University, where he also taught and Master's degree in Natural and legal rights from Aurora University. In [DATE_TIME] he made [LOCATION],

Elseid Hysaj · footballer / Coppa Italia

[PERSON] **Hysaj** (born [DATE_TIME]) is an [NRP] professional footballer who plays as a right back for [NRP] club [LOCATION] and the [LOCATION] national team. Being born in [DATE_TIME] he is also still eligible to represent the under-21 at the [DATE_TIME] UEFA European Under-21 Championship. He is an attack minded full back whose natural position is at right back, but is also comfortable playing at left back. He is an attacking fullback who is good at tracking back when faced with being caught out of position, as well as possessing good vision in order to pick out crosses as well as long range passes. [NRP] was born in [LOCATION] and started youth career with [PERSON], a



What SHAP measures

How much does token x_i **increase re-identification confidence**? Its marginal contribution to the model's prediction, averaged over all possible subsets.

Risk ranking

- Bright red: this word alone exposes the person
- Blue / grey: removing it changes very little

- Every token gets a risk score $r_i \in \mathbb{R}$

Example

Highest-risk tokens fill the anonymization queue first.

Given global budget ε and risk scores r_i :

Softmax allocation

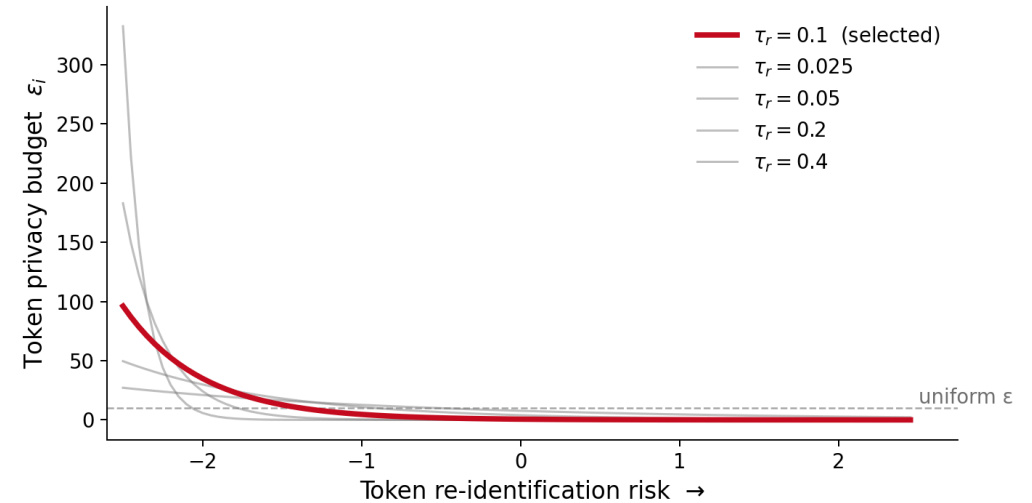
$$w_i \propto \exp(-r_i/\tau_r) \quad , \quad \sum_i w_i = 1$$

$$\varepsilon_i = \varepsilon \cdot n \cdot w_i$$

Temperature $\tau_r = 0.1$ concentrates budget on highest-risk tokens.

- **Budget preserved:** $\mathbb{E}[\varepsilon_i] = \varepsilon$
- Presidio hard-removes direct identifiers first

Risk-weighted budget allocation



- Risk-ranked tokens then rewritten under per-token ε_i

Same aggregate privacy cost as uniform — redistributed.

k-DP-MLM-X — exhaustive

Stop once the record is k-anonymous.

After **every substitution step**, the re-identification model re-evaluates the record. Anonymization halts the moment the true author drops to rank $\geq k$.

- Adapts to each record individually
- Easy records stop after very few substitutions

ρ -DP-MLM-X — simplified

Stop once cumulative residual risk $< \rho$.

Risk scores are pre-computed **once** from the initial SHAP pass. The top-risk token fraction is replaced until the remaining score sum falls below ρ .

- Uniform fraction applied across all records
- No re-evaluation during anonymization — fast

- Slower: one TRI inference per substitution step

Example

Use case: compliance — provable per-record privacy guarantee.

- Lower $\rho \rightarrow$ more tokens rewritten

Example

Use case: dataset release — fast, consistent corpus anonymization.

Results

The core question

After anonymization, can a machine learning model still identify the author of the text?

We train a re-identification model on the original documents.
It ranks all candidates. We measure where the true author lands.

- **MRR** — re-identification risk (lower = better)

Two datasets

TAB — 1 268 ECHR court judgments

Long legal documents.

Who is this applicant?

DB-Bio — 2 419 Wikipedia biographies

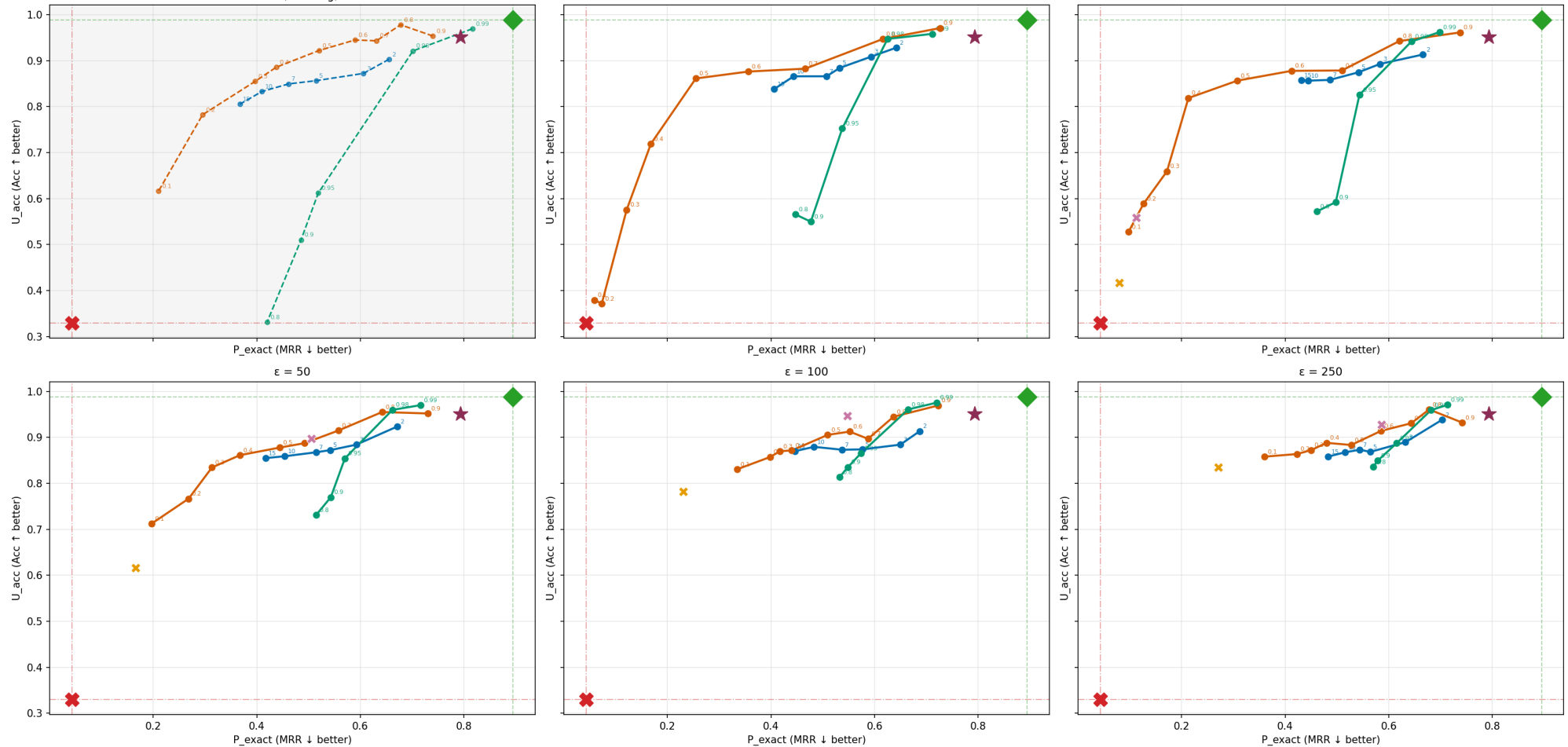
Shorter personal texts.

Who is this person?

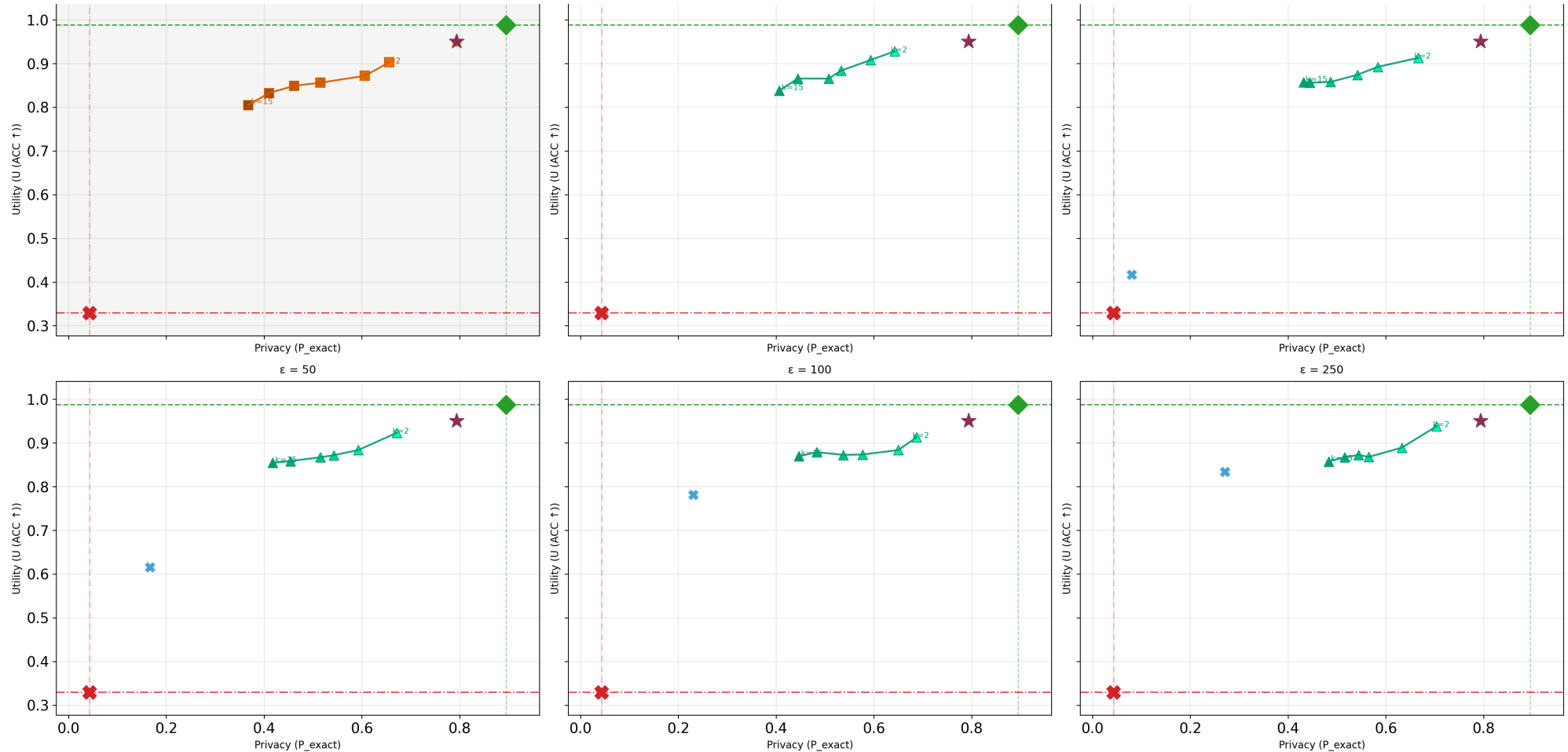
- **U** — downstream task utility (higher = better)
- **Relative gain** — single combined improvement over baseline

Baselines: entity maskers (SpaCy, Presidio),
threshold maskers (PETRE),
DP rewriters (DP-MLM uniform, DP-BART, DP-Paraphrase)

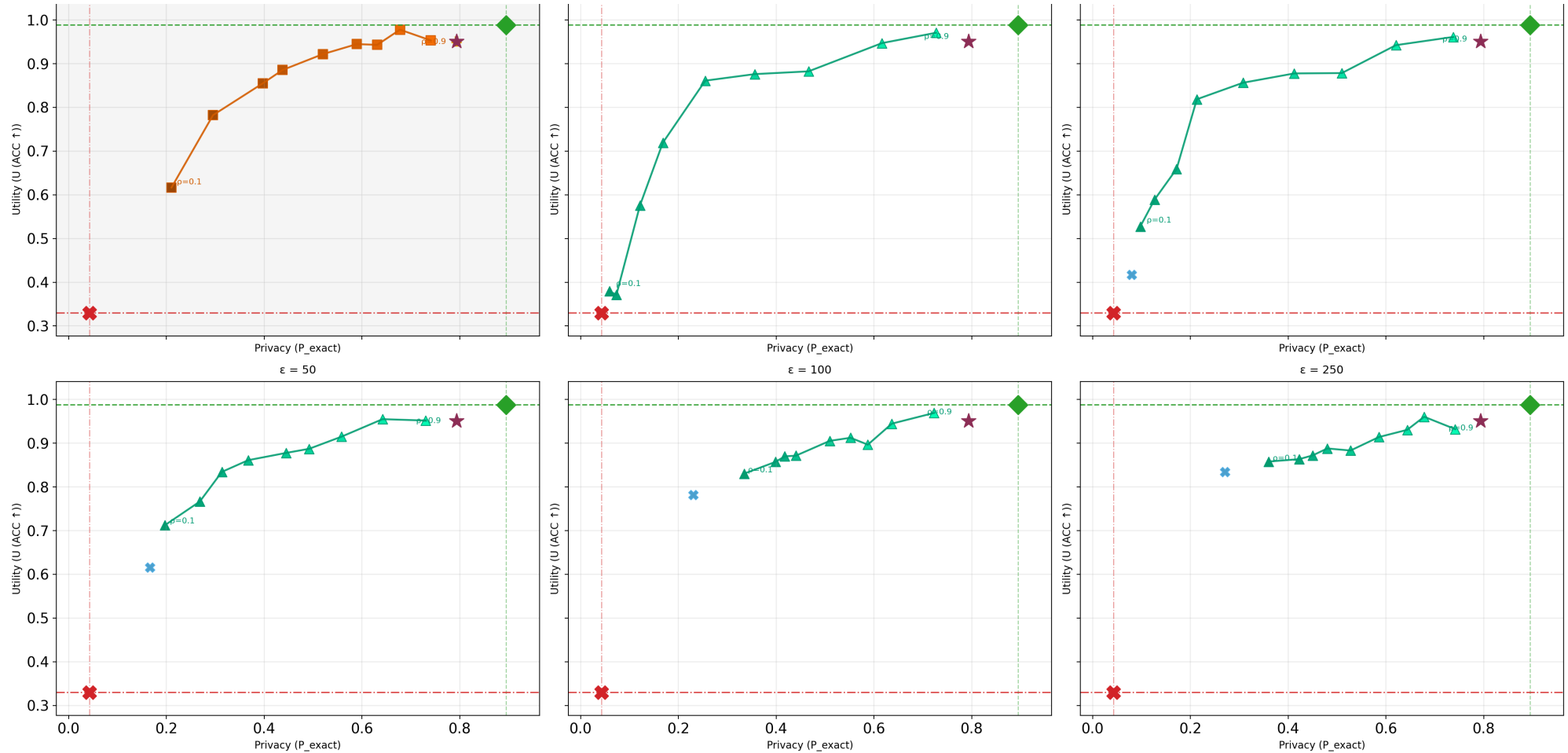
Risk-Guided Allocation Outperforms Uniform DP-MLM



k-DP-MLM-X: Full Privacy Range, Fewest Substitutions



ρ -DP-MLM-X: Best Privacy-Utility Trade-off



Takeaway

Design principle of DP-MLM-X

Concentrate DP noise where it reduces re-identification risk most.
Stop the moment the privacy criterion is satisfied.

Both variants share a **common privacy-utility space** —
the same operating points are reachable.

Practical use

Both k and ρ can be calibrated against a target privacy level before deployment. A parameter sweep maps directly to MRR and U — no guesswork.

One SHAP pass.
One parameter to tune.
Formal DP guarantee preserved.

Code: <https://anonymous.4open.science/r/risk-aware-dp>

k-DP-MLM-X exhaustive · hard per-record k-anonymity guarantee

ρ -DP-MLM-X simplified · best relative gain · fast corpus-wide anonymization

Thank You — Questions?

Yamaç Eren Ay

Faculty of Electrical Engineering and Computer Science
Technische Universität Berlin

DP-MLM-X: Risk-Aware Token-Level Differential Privacy for Text Anonymization